Bioinformatics algorithms

Rosario M. Piro
Politecnico di Milano

Multiple sequence alignment – exercises

---

1. **Multiple sequence alignment**

   Given the following MSA:

```
-  A  G  G  C  T  A  T  C  A  C  C  T  G
T  A  G  -  C  T  A  C  C  A  -  -  -  G
C  A  G  -  C  T  A  C  C  A  -  -  -  G
C  A  G  -  C  T  A  T  C  A  C  -  G  G
C  A  G  -  C  T  A  T  C  G  C  -  G  G
```

   and the following simple scoring system:

   match score = 1; mismatch penalty = -1; gap penalty = -2

   (a) Compute the total alignment score for the individual columns (1-based position in the alignment) listed in the following table. How many scores do you have to sum for each column?

Column 1:	
Column 4:	
Column 8:	

   (b) Build the alignment profile for this alignment

   (c) Align the following sequence to the existing MSA:[1] `CAGCTTTCACG`

2. **Center star algorithm**

   Given the following four DNA sequences

   (1) `ACTGTA`
   (2) `ACTTA`
   (3) `ATCTA`
   (4) `AGTGTA`

   and the following simple scoring system:

   match score = 1; mismatch penalty = -1; gap penalty = -2

   (a) For aligning these four sequences using the center star algorithm, which of the four sequences would you pick as the center of the star?

   (b) Build the multiple sequence alignment with this center of the star

3. **Information content of a sequence motif**

   Suppose you have obtained the following **local** multiple sequence alignment (so there are no gaps) of length w:

[1] This takes long! No need to do the full alignment, just get started! For the exam you would of course not get such a long task, but you should know how to set up the scoring matrix, initialize it and maybe calculate a few entries.

```
A   C   T   G   T   A
A   C   T   C   T   A
A   C   T   C   T   A
A   G   T   G   T   A
```

(a) Compute the alignment profile for the obtained multiple sequence alignment. This essential constitutes the sequence motif of, for example, a transcription factor ...

(b) Compute the information content $IC(P)$ of this alignment profile/motif P